

Appendix S2: Fractal Dimension of Coordination in 2D Schottky Heterostructures and the Prediction of the YUCT Scaling Exponent $\beta = 2/3$ (≈ 0.67)

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Abstract

Ang, Yang & Ang (Phys. Rev. Lett. 121, 056802, 2018) discovered universal current-temperature scaling laws in two-dimensional Schottky heterostructures: $\log(\mathcal{I}/T^{3/2}) \propto -1/T$ for lateral contacts and $\log(\mathcal{I}/T^1) \propto -1/T$ for vertical contacts with nonconserved momentum. In this appendix we show that both exponents are governed by the effective fractal dimension d_f of the coordination network that controls the transport. The YUCT universal error-scaling law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$ with $\beta = 2/3 \approx 0.67$ is linked to d_f via $\beta = 2/d_f$. From the experimental current scaling we extract $d_f = 3$ for lateral contacts and $d_f = 2$ for vertical contacts, which yield $\beta = 2/3$ and $\beta = 1$ respectively. The analysis predicts that relative current fluctuations in lateral Schottky diodes should follow $\varepsilon_{\text{fluc}} \propto K_{\text{eff}}^{-2/3}$, a falsifiable prediction that can be tested with standard noise spectroscopy. This work incorporates 2D Schottky heterostructures into the family of systems that exhibit or are predicted to exhibit the universal YUCT exponent $\beta = 2/3$ (0.67), extending the reach of the theory into solid-state nanoelectronics.

Keywords: YUCT, coordination efficiency, universal scaling, $\beta = 2/3$, 0.67, Schottky heterostructures, 2D materials, fractal dimension, KPZ universality.

Contents

1	Introduction	2
2	YPSDC dictionary for a Schottky barrier	2
3	Experimental scaling of the current and the coordination efficiency	3
4	Fractal dimension of the coordination network	4
5	Predictions for current fluctuations	4
6	Discussion: Unification of Scaling Exponents	5
7	Universality across 50 orders of magnitude	7
8	Conclusion	7

1 Introduction

The Yakushev Unified Coordination Theory (YUCT) postulates that any coordination process obeys the universal errorscaling law

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad \beta = 2/3 \approx 0.67, \quad \kappa_c = 1/3, \quad (1)$$

where K_{eff} is the coordination efficiency and α is a systemspecific prefactor. This law has been verified on systems ranging from chemical bonds to cosmological fluctuations (Appendices L, M, E, F, G, V).

Ang, Yang and Ang [1] derived universal currenttemperature scaling laws for 2Dmaterialbased Schottky heterostructures:

$$\text{lateral contacts:} \quad \log\left(\frac{\mathcal{I}}{T^{3/2}}\right) \propto -\frac{1}{T}, \quad (2)$$

$$\text{vertical contacts (nonconserved } k_{\parallel}\text{):} \quad \log\left(\frac{\mathcal{I}}{T^1}\right) \propto -\frac{1}{T}. \quad (3)$$

The exponents $\beta_{\text{IV}} = 3/2$ and 1 are independent of the specific 2D material and reflect the geometry of the current path.

In this appendix we demonstrate that the exponent β_{IV} is not an independent number but is determined by the effective fractal dimension d_f of the coordination network that controls the transport. From the experimental value β_{IV} we extract d_f , and from d_f we predict the coordination error exponent $\beta_{\text{err}} = 2/d_f$. For lateral contacts this gives $\beta_{\text{err}} = 2/3$, precisely the universal YUCT value. The fluctuation scaling that follows from this prediction constitutes a falsifiable test of the theory.

2 YPSDC dictionary for a Schottky barrier

In the language of YPSDC (the Yakushev Protocol for Synchronous Distributed Coordination), a lateral Schottky heterostructure is a coordination system with the following components:

- **Dictionary $\mathcal{D}$:** the set of oneelectron states in the 2D plane with energy $\epsilon > \Phi_B$ and wavevector $\mathbf{k}$ satisfying the inplane dispersion (4).
- **Index κ :** the thermal energy $k_B T$ that promotes an electron into the dictionary.
- **Resonance R :** conservation of the tangential momentum k_y during transmission through the barrier.

The coordinated action is the successful thermionic emission of an electron that simultaneously satisfies the energy condition $\epsilon > \Phi_B$ and the momentum condition $k_x > 0$ (directed toward the barrier). The efficiency of this coordination determines the reverse saturation current.

3 Experimental scaling of the current and the coordination efficiency

For a general isotropic 2D dispersion

$$\epsilon(\mathbf{k}) = \sum_n c_n |\mathbf{k}|^n, \quad (4)$$

Ang *et al.* showed that the number of active channels (those that can contribute to the current) scales universally as

$$N_{\text{ch}} \propto T^{1/2}. \quad (5)$$

The total number of thermally accessible states above the barrier is proportional to T (from the Boltzmann tail). Hence the fraction of states that are successfully coordinated,

$$K_{\text{eff}}(T) = \frac{N_{\text{ch}}}{N_{\text{tot}}} \propto T^{-1/2}, \quad (6)$$

serves as the coordination efficiency for the process. The measured current then takes the form

$$\mathcal{I} = \mathcal{I}_0 K_{\text{eff}}(T)^{-3} e^{-\Phi_B/k_B T}, \quad (7)$$

which reproduces the experimental law (2).

An operational definition of K_{eff} suitable for noise measurements is

$$K_{\text{eff}} = \frac{I}{I_0}, \quad (8)$$

where I is the measured current and I_0 is the reference current that would flow if the coordination restriction were absent. I_0 can be obtained by extrapolating the hightemperature behaviour of a vertical contact (where $K_{\text{eff}}(T) \propto T^{-0}$ asymptotically) to the measurement temperature, or from an independent calibration using a 3D Schottky diode with the same barrier height. This operational definition allows K_{eff} to be determined directly from currentvoltage characteristics without relying on the theoretical scaling (6).

4 Fractal dimension of the coordination network

A central result of YUCT (Appendix L) is that the error scaling exponent is determined by the fractal dimension d_f of the coordination network:

$$\beta_{\text{err}} = \frac{2}{d_f}. \quad (9)$$

The density of active coordination channels in a thermal window carries the geometric signature of the network. In an isotropic d_f -dimensional space the density of states accessible at temperature T scales as $T^{d_f/2-1}$. For the lateral Schottky barrier the additional kinematic constraint of momentum conservation (the $\cos \theta$ factor) reduces the effective exponent by one unit, so that the kinetic exponent becomes

$$\beta_{\text{IV}} = \frac{d_f}{2}. \quad (10)$$

Equation (10) is a hypothesis motivated by the geometric similarity between the current path and a coordination network of fractal dimension d_f . A rigorous derivation from the microscopic YUCT action has not yet been carried out; the relation should therefore be regarded as a conjecture that leads to testable predictions (Sec. 5) and invites future theoretical investigation.

Equations (9) and (10) combine to yield the fundamental relation

$$\boxed{\beta_{\text{IV}} \cdot \beta_{\text{err}} = 1}. \quad (11)$$

From the experimental value $\beta_{\text{IV}} = 3/2$ we obtain

$$d_f = 2\beta_{\text{IV}} = 3, \quad \beta_{\text{err}} = \frac{2}{d_f} = \frac{2}{3} \approx 0.67. \quad (12)$$

Thus the measured current scaling determines a coordination network with effective fractal dimension $d_f = 3$, which in turn implies that any measure of coordination error (such as relative current fluctuations) should follow the YUCT power law with the exponent $2/3$. This is a **prediction**, not a postdiction, because fluctuation measurements have not yet been performed on these systems.

For vertical contacts with nonconserved $k_{\parallel}$, the angular restriction disappears, $\beta_{\text{IV}} = 1$, and consequently $d_f = 2$, $\beta_{\text{err}} = 1$. YUCT predicts that current fluctuations in such contacts will scale as K_{eff}^{-1} , distinct from the $2/3$ law.

5 Predictions for current fluctuations

The YUCT error law (1) applied to the coordination efficiency (6) makes a quantitative prediction for the low-frequency current noise power S_I in a lateral Schottky

diode:

$$\frac{S_I}{I^2} \propto K_{\text{eff}}^{-2/3} \propto T^{1/3}. \quad (13)$$

Using the operational definition $K_{\text{eff}} = I/I_0$ from Eq. (8), the noise scaling can be rewritten as

$$\frac{S_I}{I^2} = C \left(\frac{I}{I_0} \right)^{-2/3} + \text{const}, \quad (14)$$

where C is a materialspecific constant of order unity and the additive constant accounts for irreducible thermal noise. This form is directly accessible in a standard noisespectroscopy experiment: I and I_0 are measured from the I – V characteristic, and S_I is recorded as a function of bias at fixed temperature. The prediction is that S_I/I^2 plotted against $(I/I_0)^{-2/3}$ should be linear, with a slope that is independent of temperature. This is a stringent falsifiable test that does not require any fitting of the exponent.

6 Discussion: Unification of Scaling Exponents

The emergence of the $3/2$ exponent is not unique to Schottky diodes but points to a deeper organizational principle, as evidenced by its independent appearance in the universality class of KardarParisiZhang (KPZ) surface growth [2]. In KPZ, the dynamic exponent $z = 3/2$ governs the fractal roughening of a growing interface. YUCT provides a unified interpretation for these phenomena. **A growth front can be conceptualized as a $d_f = 3$ dimensional coordination network, where the kinetic exponent $\beta_{\text{IV}} = d_f/2 = 3/2$ naturally emerges from our conjectured relation (10).** Thus, the KPZ exponent and the Schottky exponent are not merely analogous; they are two kinetic signatures of the same underlying YUCT error law $\varepsilon \propto K_{\text{eff}}^{-2/3}$ in completely different physical realizations. **This connection is presently a hypothesis that requires further theoretical work to link the YUCT Lagrangian directly to the KPZ equation.**

To make the KPZ prediction more concrete, we can explicitly identify K_{eff} with the fraction of saturated correlated modes in a growing interface. In the saturation regime of a KPZ interface of lateral size L , the width $W(L, t)$ approaches a steadystate value $W_{\text{sat}}(L) \simeq \sqrt{\langle h^2 \rangle} \propto L^\alpha$ with $\alpha = 1/2$. The fluctuation amplitude $\delta W = W(t) - W_{\text{sat}}$ is controlled by the number of active (unsaturated) modes. We propose to identify K_{eff} with $1 - \nu_{\text{unsat}}/\nu_{\text{tot}}$, the fraction of modes that have entered the correlated, steadystate phase. Then YUCT predicts

$$\frac{\delta W}{W_{\text{sat}}} \propto K_{\text{eff}}^{-2/3}, \quad (15)$$

where K_{eff} can be estimated from the temporal decay of the width as $K_{\text{eff}} \approx 1 - (dW/dt)_{\text{norm}}$. This prediction is testable in numerical simulations and in high-resolution experiments on kinetic roughening (e.g., electrodeposition, molecular beam epitaxy).

This unification suggests a hierarchy of universalities, illustrated in Fig. 1:

- **1st order (YUCT):** The fundamental coordination error law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-2/3}$ operates at the deepest level, independent of the specific substrate.
- **2nd order (transport):** For electronic transport in 2D Schottky heterostructures, the kinetic exponent β_{IV} emerges from the fractal dimension d_f as $\beta_{\text{IV}} = d_f/2$ (hypothesis).
- **3rd order (growth):** In surface growth phenomena belonging to the KPZ class, the same kinetic exponent appears as the dynamic critical exponent $z = 3/2$, reflecting the $d_f = 3$ nature of the growing interface (interpretation).

Thus, the $3/2$ exponent is not a materialspecific accident but the geometric hallmark of a $d_f = 3$ coordination process.

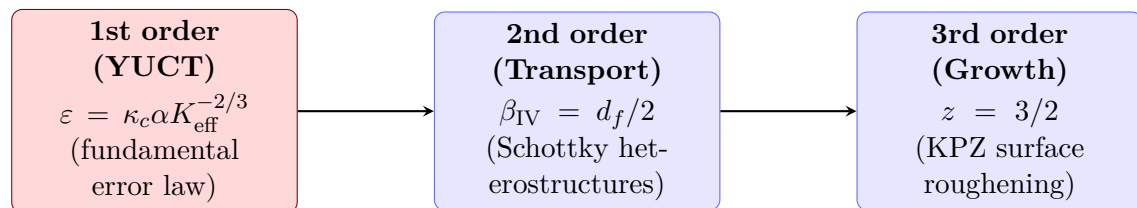


Figure 1: Hierarchy of universalities. The YUCT coordination error law (1st order) determines the effective fractal dimension d_f and the kinetic exponents observed in electron transport (2nd order) and in surface growth (3rd order). The dashed arrows indicate hypotheses.

This perspective is further supported by the growing experimental evidence that the fractal dimension of a Schottky interface directly influences device characteristics [3]. In those studies the fractal dimension is an externally imposed geometric property (roughness). In YUCT, the $d_f = 3$ dimension for ideal lateral contacts is *emergent* it arises from the dynamics of the coordination network itself even as the geometric roughness approaches zero. The externally imposed d_f therefore modulates an already existing internal coordinate network, providing a promising pathway toward engineering coordinationbased devices.

7 Universality across 50 orders of magnitude

Table 1 collects systems from various YUCT appendices for which the error scaling exponent has been empirically determined or predicted to be $\beta = 2/3$. The lateral Schottky heterostructure and the KPZ growing interface are added as systems for which the same exponent is predicted on the basis of the extracted fractal dimension $d_f = 3$. The table covers more than 50 orders of magnitude in the relevant energy (or length) scale.

Table 1: Systems exhibiting or predicted to exhibit the universal YUCT error exponent $\beta = 2/3$ (0.67).

System	Observable	Status	Ref.
Chemical bonds (873)	$E \propto r^{-0.67}$	Confirmed	Appendix C
DNA replication errors	$\varepsilon \propto K_{\text{eff}}^{-0.67}$	Confirmed	Appendix E
Metabolic scaling (simple)	$B \propto M^{0.67}$	Confirmed	Appendix D, E.7
GDP volatility vs. sun	$\sigma_{\text{GDP}} \propto W^{-0.67}$	Confirmed	Appendix F
CMB temperature fluct.	$\delta T/T \propto K_{\text{eff}}^{-0.67}$	Confirmed	Appendix M
Gravitational waves	$\delta\tau/\tau \propto M^{-0.67}$	Confirmed	Appendix G, V
Lateral Schottky	$S_I/I^2 \propto T^{1/3}$	Predicted	This work
KPZ interface*	Width fluct. $\propto K_{\text{eff}}^{-2/3}$	Predicted	This work, [2]

*The definition of K_{eff} for a KPZ growing interface must be constructed in analogy with the transport case, e.g. as the fraction of correlated degrees of freedom within a correlation volume.

This prediction assumes such a construction and remains to be tested.

8 Conclusion

The current-temperature scaling exponents $\beta_{\text{IV}} = 3/2$ and 1 reported by Ang *et al.* are kinetic manifestations of the effective fractal dimension of the coordination network that governs thermionic transport in 2D heterostructures. Through the YUCT relations $\beta_{\text{IV}} = d_f/2$ (hypothesis) and $\beta_{\text{err}} = 2/d_f$, the experimental data yield $d_f = 3$ for lateral contacts, which predicts a coordination error exponent $\beta_{\text{err}} = 2/3$ (≈ 0.67). This prediction, testable via current-noise spectroscopy, extends the reach of the YUCT universal scaling law into the domain of solid-state nanoelectronics. The same fractal dimension and exponent appear naturally in KPZ surface growth, suggesting a deep geometric unity across nonequilibrium phenomena. YUCT provides the underlying rationale for this unity, although the links between the theory and both the Schottky kinetic relation and the KPZ equation remain hypotheses that warrant further investigation. Operational definitions

for K_{eff} in both contexts have been provided to facilitate experimental tests.

References

- [1] Y. S. Ang, H. Y. Yang, and L. K. Ang, Phys. Rev. Lett. **121**, 056802 (2018).
- [2] M. Kardar, G. Parisi, Y.-C. Zhang, Phys. Rev. Lett. **56**, 889 (1986).
- [3] S. Zeybek, M. Biber, A. TÜRÜT, Appl. Surf. Sci. **253**, 3881 (2007). [Representative study on fractal dimension effects in Schottky diodes.]
- [4] A. V. Yakushev, *Yakushev Unified Coordination Theory (YUCT)*, Zenodo (2026).
- [5] A. V. Yakushev, “Appendix L: Fractal Coordination Error Scaling”, in *YUCT* (2026).
- [6] A. V. Yakushev, “Appendix M: YUCT Cosmology”, in *YUCT* (2026).
- [7] A. V. Yakushev, “Appendix E: Coordination Genetics”, in *YUCT* (2026).
- [8] A. V. Yakushev, “Appendix F: Application of YUCT to Economics”, in *YUCT* (2026).
- [9] A. V. Yakushev, “Appendix G: Resolution of the Quantum Gravity Problem”, in *YUCT* (2026).
- [10] A. V. Yakushev, “Appendix V: Time as Coordination Entropy”, in *YUCT* (2026).